

گزارش تمرین پنجم

تشخیص وسایل نقلیه با خانواده YOLO و تولید خودکار برچسب‌ها با YOLOv8

کاری از: سارا محمدی

شرح مسئله

هدف، آماده‌سازی یک مجموعه داده تشخیص شیء برای پنج کلاس Car، Bicycle، Bus، Truck و Motorcycle، تولید برچسب‌های اولیه، تقسیم داده به آموزش و اعتبارسنجی و آزمون، آموزش مدل‌های مختلف YOLO و مقایسه دقت آن‌ها است.

نکته درباره نسخه مدل

در نوت‌بوک، مرحله Teacher و تولید برچسب‌ها با وزن yolov5su.pt انجام شده است. بخش آموزش مقایسه‌ای ذخیره شده در همین فایل با YOLOv8s، YOLOv8m و YOLOv8l اجرا شده؛ بنابراین نتایج عددی و نمودارهای گزارش متعلق به اجرای واقعی همان سه مدل هستند و بدون تغییر درج شده‌اند.

۱. نصب کتابخانه‌ها

```
!pip -q install ultralytics comet_ml reportlab pyyaml gdown
```

```
----- 45.5/45.5 kB 4.1 MB/s eta 0:00:00
----- 1.4/1.4 MB 18.5 MB/s eta 0:00:00
----- 796.6/796.6 kB 11.2 MB/s eta 0:00:00
----- 2.0/2.0 MB 26.8 MB/s eta 0:00:00
----- 1.5/1.5 MB 16.5 MB/s eta 0:00:00
----- 63.9/63.9 kB 2.8 MB/s eta 0:00:00
```

۲. کتابخانه‌ها و بررسی GPU

```
import comet_ml

import gc
import os
import random
import shutil
import time
import zipfile
from pathlib import Path

import cv2
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import torch
import yaml
from PIL import Image as PILImage
from ultralytics import YOLO

random.seed(0)
np.random.seed(0)
print('GPU:', torch.cuda.get_device_name(0) if torch.cuda.is_available() else 'no GPU')
```

```
Creating new Ultralytics Settings v0.0.7 file â€¦
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see https://docs.ultralytics.com/quickstart/#ultralytics-GPU: Tesla T4
```

۳. دریافت مجموعه تصاویر وسایل نقلیه

```
import gdown

FILE_ID = '1WrBTlnxo__PSUTuL06sIHaxf9apQMccJ'
ZIP_PATH = '/content/vehicle-images.zip'
RAW_DIR = Path('/content/vehicle_raw')

gdown.download(id=FILE_ID, output=ZIP_PATH, quiet=False)

with zipfile.ZipFile(ZIP_PATH) as archive:
    archive.extractall(RAW_DIR)

extensions = {'.jpg', '.jpeg', '.png', '.bmp'}
raw_images = sorted(p for p in RAW_DIR.rglob('*') if p.suffix.lower() in extensions)
print('photo Num :', len(raw_images))
```

```
Downloading...
From (original): https://drive.google.com/uc?id=1WrBTlnxo\_\_PSUTuL06sIHaxf9apQMccJ
From (redirected): https://drive.google.com/uc?id=1WrBTlnxo\_\_PSUTuL06sIHaxf9apQMccJ&confirm=t&uuid=19d5a7e2-7663-4dbc-a446-316c205506d9
To: /content/vehicle-images.zip
100%|██████████| 124M/124M [00:03<00:00, 39.0MB/s]
photo Num : 100
```

۴. تبدیل تصاویر به قالب JPEG

```
CONVERTED_DIR = Path('/content/vehicle_converted')
CONVERTED_DIR.mkdir(exist_ok=True)

converted_images = []
for i, source in enumerate(raw_images, start=1):
    dest = CONVERTED_DIR / f'vehicle_{i:04d}.jpg'
    with PILImage.open(source) as img:
        img.convert('RGB').save(dest, 'JPEG', quality=92)
    converted_images.append(dest)

print('Number of photos converted:', len(converted_images))
```

```
Number of photos converted: 100
```

همه تصاویر با نام‌گذاری یکنواخت به JPEG RGB تبدیل می‌شوند تا ورودی پایدار و قابل پردازش باشد.

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۵. تولید برچسب‌ها با YOLOv5

```
CLASS_NAMES = ['Car', 'Bicycle', 'Bus', 'Truck', 'Motorcycle']
COCO_TO_CUSTOM = {2: 0, 1: 1, 5: 2, 7: 3, 3: 4}

IMAGES_DIR = Path('/content/labeled/images')
LABELS_DIR = Path('/content/labeled/labels')
IMAGES_DIR.mkdir(parents=True, exist_ok=True)
LABELS_DIR.mkdir(parents=True, exist_ok=True)

teacher = YOLO('yolov5su.pt')
results = teacher.predict(
    source=[str(p) for p in converted_images],
    conf=0.2, iou=0.5, imgsz=640,
    device=0 if torch.cuda.is_available() else 'cpu',
    stream=True, verbose=False
)

object_counts = np.zeros(len(CLASS_NAMES), dtype=int)
for result in results:
    source = Path(result.path)
    shutil.copy2(source, IMAGES_DIR / source.name)

    lines = []
    if result boxes is not None:
        for coco_id, box in zip(result.boxes.cls.cpu().numpy().astype(int), result.boxes.xywhn.cpu().numpy()):
            if coco_id in COCO_TO_CUSTOM:
                cls_id = COCO_TO_CUSTOM[coco_id]
                x, y, w, h = box
                lines.append(f'{cls_id} {x:.6f} {y:.6f} {w:.6f} {h:.6f}')
                object_counts[cls_id] += 1
    (LABELS_DIR / f'{source.stem}.txt').write_text('\n'.join(lines))

del teacher
gc.collect()
torch.cuda.empty_cache()

print(pd.DataFrame({'Class': CLASS_NAMES, 'Objects': object_counts}))
```

Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov5su.pt> to 'yolov5su.pt': 100% — 17.7MB 320.6MB/s 0.1s

	Class	Objects
0	Car	589
1	Bicycle	0
2	BUS	6
3	Truck	32
4	Motorcycle	4

مدل yolov5su.pt نقش Teacher را دارد. پیش‌بینی‌ها به پنج کلاس سفارشی نگاشت شده و در قالب استاندارد YOLO در فایل‌های متنی ذخیره می‌شوند. حالت stream از انباشته شدن همه نتایج در حافظه جلوگیری می‌کند.

۶. نمایش نمونه برچسب‌ها

```
COLORS = [(0, 200, 0), (255, 120, 0), (0, 120, 255), (180, 0, 180), (255, 0, 0)]

def draw_labels(image_path, label_path):
    image = cv2.cvtColor(cv2.imread(str(image_path)), cv2.COLOR_BGR2RGB)
    h, w = image.shape[:2]
    if label_path.exists() and label_path.stat().st_size:
        for line in label_path.read_text().splitlines():
            cls_id, x, y, bw, bh = map(float, line.split())
            cls_id = int(cls_id)
            x1, y1 = int((x - bw / 2) * w), int((y - bh / 2) * h)
            x2, y2 = int((x + bw / 2) * w), int((y + bh / 2) * h)
            color = COLORS[cls_id]
            cv2.rectangle(image, (x1, y1), (x2, y2), color, 2)
            cv2.putText(image, CLASS_NAMES[cls_id], (x1, max(y1 - 5, 15)), cv2.FONT_HERSHEY_SIMPLEX, 0.6, color, 2)
    return image

sample_paths = random.sample(sorted(IMAGES_DIR.glob '*.jpg'), 12)
fig, axes = plt.subplots(3, 4, figsize=(16, 10))
for ax, path in zip(axes.ravel(), sample_paths):
    ax.imshow(draw_labels(path, LABELS_DIR / f'{path.stem}.txt'))
    ax.set_title(path.name, fontsize=9)
    ax.axis('off')
plt.tight_layout()
plt.show()
```



چند تصویر انتخابی همراه با کادر، نام کلاس و شناسه برچسب نمایش داده شده‌اند تا کیفیت برچسب‌گذاری کنترل شود.

۷. تقسیم داده و ساخت فایل YAML

```
DATASET_DIR = Path('/content/vehicle_dataset')
if DATASET_DIR.exists():
    shutil.rmtree(DATASET_DIR)
for split in ['train', 'val', 'test']:
    (DATASET_DIR / 'images' / split).mkdir(parents=True, exist_ok=True)
    (DATASET_DIR / 'labels' / split).mkdir(parents=True, exist_ok=True)

all_images = sorted(IMAGE_DIR.glob('*.jpg'))
random.Random(0).shuffle(all_images)
n_train = int(0.70 * len(all_images))
n_val = int(0.15 * len(all_images))
splits = {
    'train': all_images[:n_train],
    'val': all_images[n_train:n_train + n_val],
    'test': all_images[n_train + n_val:],
}
for split, paths in splits.items():
    for image_path in paths:
        shutil.copy2(image_path, DATASET_DIR / 'images' / split / image_path.name)
        label_path = LABELS_DIR / f'{image_path.stem}.txt'
        shutil.copy2(label_path, DATASET_DIR / 'labels' / split / label_path.name)
    print(f'{split}: {len(paths)} photos')

yaml_data = {
    'path': str(DATASET_DIR.resolve()),
    'train': 'images/train',
    'val': 'images/val',
    'test': 'images/test',
    'names': {i: name for i, name in enumerate(CLASS_NAMES)},
}
YAML_PATH = DATASET_DIR / 'vehicle.yaml'
YAML_PATH.write_text(yaml.safe_dump(yaml_data, sort_keys=False))
print(YAML_PATH.read_text())
```

```
train: 70 photos
val: 15 photos
test: 15 photos
path: /content/vehicle_dataset
train: images/train
val: images/val
test: images/test
names:
  0: Car
  1: Bicycle
  2: Bus
  3: Truck
  4: Motorcycle
```

داده‌ها با نسبت ۷۰ درصد آموزش، ۱۵ درصد اعتبارسنجی و ۱۵ درصد آزمون تقسیم شده و فایل data.yaml شامل

مسیرها و نام پنج کلاس ساخته می‌شود.

۸. اتصال اختیاری Comet

```
os.environ['COMET_MODE'] = 'disabled'
os.environ['COMET_PROJECT_NAME'] = 'vehicle-yolov8'
from google.colab import userdata
try:
    key = userdata.get('COMET_API_KEY')
    comet_ml.login(api_key=key)
    os.environ['COMET_MODE'] = 'online'
    print('Comet Connected.')
except Exception:
    print('Comet failed to connect,')
```

```
[1;38;5;39mCOMET INFO:[0m Valid Comet API Key saved in /root/.comet.config (set COMET_CONFIG to change where it is saved).
```

```
Comet Connected.
```

اتصال به Comet در صورت وجود کلید API فعال می‌شود؛ در غیر این صورت ثبت آزمایش غیرفعال باقی می‌ماند.

۹. آموزش و ارزیابی سه اندازه مدل

```
MODEL_SIZES = ['s', 'm', 'l']
EPOCHS = 60
IMG_SIZE = 640
BATCH_SIZE = 8
DEVICE = 0 if torch.cuda.is_available() else 'cpu'

results_rows = []
model_paths = {}

for size in MODEL_SIZES:
    name = f'yolov8{size}'
    print(f'\n== {name} ==')

    model = YOLO(f'{name}.pt')
    start = time.time()
    model.train(
        data=str(YAML_PATH), epochs=EPOCHS, imgsz=IMG_SIZE, batch=BATCH_SIZE,
        patience=15, optimizer='AdamW', lr0=1e-3, device=DEVICE, seed=0, verbose=False,
        project='vehicle_runs', name=name,
    )
    train_minutes = (time.time() - start) / 60

    best_path = Path(model.trainer.save_dir) / 'weights' / 'best.pt'
    best_model = YOLO(str(best_path))
    val_result = best_model.val(data=str(YAML_PATH), split='test', imgsz=IMG_SIZE, device=DEVICE, verbose=False)
    metrics = val_result.results_dict

    results_rows.append({
        'Model': name,
        'Precision': metrics.get('metrics/precision(B)', np.nan),
        'Recall': metrics.get('metrics/recall(B)', np.nan),
        'mAP50': metrics.get('metrics/mAP50(B)', np.nan),
        'mAP50-95': metrics.get('metrics/mAP50-95(B)', np.nan),
        'Inference_ms': val_result.speed.get('inference', np.nan),
        'Train_minutes': train_minutes,
    })
    model_paths[name] = {'best': str(best_path), 'train_dir': str(model.trainer.save_dir)}

del model, best_model
gc.collect()
torch.cuda.empty_cache()
```

```
== yolov8s ==
Ultralytics 8.4.120 ðŸš€ Python-3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
[34m[1mengine/trainer: [0magnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugmt, batch=8, bgr=0.0, box=7.5, cache=False, cfg=No

[1;38;5;39mCOMET INFO:[0m An experiment with the same configuration options is already running and will be reused.

Overriding model.yaml nc=80 with nc=5

      from  n  params  module                                arguments
0         -1  1      928  ultralytics.nn.modules.conv.Conv          [3, 32, 3, 2]
1         -1  1     18560  ultralytics.nn.modules.conv.Conv          [32, 64, 3, 2]
2         -1  1     29056  ultralytics.nn.modules.block.C2f          [64, 64, 1, True]
3         -1  1     73984  ultralytics.nn.modules.conv.Conv          [64, 128, 3, 2]
4         -1  2     197632  ultralytics.nn.modules.block.C2f          [128, 128, 2, True]
5         -1  1     295424  ultralytics.nn.modules.conv.Conv          [128, 256, 3, 2]
6         -1  2     788480  ultralytics.nn.modules.block.C2f          [256, 256, 2, True]
7         -1  1    1180672  ultralytics.nn.modules.conv.Conv          [256, 512, 3, 2]
8         -1  1    1838080  ultralytics.nn.modules.block.C2f          [512, 512, 1, True]
```

```
Remapped 5/5 cls head rows from pretrained weights by class name
Transferred 355/355 items from pretrained weights
```

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
-------	---------	----------	----------	----------	-----------	------

[illegible]

[illegible][illegible]

```
Speed: 0.1ms preprocess, 4.0ms inference, 0.0ms loss, 0.9ms postprocess per image
Results saved to [1m/content/runs/detect/vehicle_runs/yolov8s-2[0m

[1;38;5;39mCOMET INFO:[0m -----
[1;38;5;39mCOMET INFO:[0m Comet.ml Experiment Summary
[1;38;5;39mCOMET INFO:[0m -----
[1;38;5;39mCOMET INFO:[0m Data:
[1;38;5;39mCOMET INFO:[0m     display_summary_level : 1
[1;38;5;39mCOMET INFO:[0m     name                  : subtle_redshift_142
[1;38;5;39mCOMET INFO:[0m     url                   : https://www.comet.com/sarah-mohammadi/vehicle-yolov8/4a73b1c299a94f239502c42b5eafb2d0
[1;38;5;39mCOMET INFO:[0m Metrics [count] (min, max):
[1;38;5;39mCOMET INFO:[0m     loss [13]              : (35742.890625, 1978738.875)
[1;38;5;39mCOMET INFO:[0m     lr/pg0 [17]            : (0.0002962962962962963, 0.0009505000000000001)
[1;38;5;39mCOMET INFO:[0m     lr/pg1 [17]            : (0.0002962962962962963, 0.0009505000000000001)
[1;38;5;39mCOMET INFO:[0m     lr/pg2 [17]            : (0.0007524999999999999, 0.07066666666666667)
[1;38;5;39mCOMET INFO:[0m     metrics/mAP50(B) [17]  : (0.1022, 0.51703)
[1;38;5;39mCOMET INFO:[0m     metrics/mAP50-95(B) [17] : (0.0625, 0.3163380589490771)
[1;38;5;39mCOMET INFO:[0m     metrics/precision(B) [17] : (0.44695, 0.94305)
```

```

[1;38;5;39mCOMET INFO:[0m metrics/recall(B) [17] : (0.15421, 0.60278)
[1;38;5;39mCOMET INFO:[0m model/GFLOPs : 28.651
[1;38;5;39mCOMET INFO:[0m model/parameters : 11137535
[1;38;5;39mCOMET INFO:[0m model/speed_PyTorch(ms) : 11.177
[1;38;5;39mCOMET INFO:[0m train/box_loss [16] : (0.57824, 1.19995)
[1;38;5;39mCOMET INFO:[0m train/cls_loss [16] : (0.57615, 2.11327)
[1;38;5;39mCOMET INFO:[0m train/df1_loss [16] : (0.92798, 1.18594)
[1;38;5;39mCOMET INFO:[0m val/box_loss [16] : (0.64643, 1.52626)
[1;38;5;39mCOMET INFO:[0m val/cls_loss [16] : (0.8785, 5.32776)
[1;38;5;39mCOMET INFO:[0m val/df1_loss [16] : (0.97418, 1.31435)
[1;38;5;39mCOMET INFO:[0m Others:
[1;38;5;39mCOMET INFO:[0m eval_batch_logging_interval : 1
[1;38;5;39mCOMET INFO:[0m log_confusion_matrix_on_eval : False
[1;38;5;39mCOMET INFO:[0m log_image_predictions : True
[1;38;5;39mCOMET INFO:[0m max_image_predictions : 100
[1;38;5;39mCOMET INFO:[0m notebook_url : https://colab.research.google.com/notebook#fileId=1ILLavEZtYHkYpJPRrvtMnhSrtv2fne9v
[1;38;5;39mCOMET INFO:[0m Parameters:
[1;38;5;39mCOMET INFO:[0m agnostic_nms : False

```

```

[1;38;5;39mCOMET INFO:[0m amp : True
[1;38;5;39mCOMET INFO:[0m angle : 1.0
[1;38;5;39mCOMET INFO:[0m augment : False
[1;38;5;39mCOMET INFO:[0m auto_augment : randaugment
[1;38;5;39mCOMET INFO:[0m batch : 8
[1;38;5;39mCOMET INFO:[0m bgr : 0.0
[1;38;5;39mCOMET INFO:[0m box : 7.5
[1;38;5;39mCOMET INFO:[0m cache : False
[1;38;5;39mCOMET INFO:[0m cfg : None
[1;38;5;39mCOMET INFO:[0m channels_last : False
[1;38;5;39mCOMET INFO:[0m classes : None
[1;38;5;39mCOMET INFO:[0m close_mosaic : 10
[1;38;5;39mCOMET INFO:[0m cls : 0.5
[1;38;5;39mCOMET INFO:[0m cls_pw : 0.0
[1;38;5;39mCOMET INFO:[0m cls_remap : True
[1;38;5;39mCOMET INFO:[0m compile : False
[1;38;5;39mCOMET INFO:[0m conf : None
[1;38;5;39mCOMET INFO:[0m copy_paste : 0.0

```

```

[1;38;5;39mCOMET INFO:[0m copy_paste_mode : flip
[1;38;5;39mCOMET INFO:[0m cos_lr : False
[1;38;5;39mCOMET INFO:[0m cutmix : 0.0
[1;38;5;39mCOMET INFO:[0m data : /content/vehicle_dataset/vehicle.yaml
[1;38;5;39mCOMET INFO:[0m degrees : 0.0
[1;38;5;39mCOMET INFO:[0m deterministic : True
[1;38;5;39mCOMET INFO:[0m device : 0
[1;38;5;39mCOMET INFO:[0m df1 : 1.5
[1;38;5;39mCOMET INFO:[0m dgrad : 0.5
[1;38;5;39mCOMET INFO:[0m dis : 6.0
[1;38;5;39mCOMET INFO:[0m distill_model : None
[1;38;5;39mCOMET INFO:[0m dlam : 1.0
[1;38;5;39mCOMET INFO:[0m dlog : 1.0
[1;38;5;39mCOMET INFO:[0m dnn : False
[1;38;5;39mCOMET INFO:[0m dropout : 0.0
[1;38;5;39mCOMET INFO:[0m dynamic : False
[1;38;5;39mCOMET INFO:[0m embed : None
[1;38;5;39mCOMET INFO:[0m end2end : None

```

```

[1;38;5;39mCOMET INFO:[0m epochs : 60
[1;38;5;39mCOMET INFO:[0m erasing : 0.4
[1;38;5;39mCOMET INFO:[0m exist_ok : False
[1;38;5;39mCOMET INFO:[0m fliplr : 0.5
[1;38;5;39mCOMET INFO:[0m flipud : 0.0
[1;38;5;39mCOMET INFO:[0m format : torchscript
[1;38;5;39mCOMET INFO:[0m fraction : 1.0
[1;38;5;39mCOMET INFO:[0m freeze : None
[1;38;5;39mCOMET INFO:[0m hsv_h : 0.015
[1;38;5;39mCOMET INFO:[0m hsv_s : 0.7
[1;38;5;39mCOMET INFO:[0m hsv_v : 0.4
[1;38;5;39mCOMET INFO:[0m imsz : 640
[1;38;5;39mCOMET INFO:[0m iou : 0.7
[1;38;5;39mCOMET INFO:[0m keras : False
[1;38;5;39mCOMET INFO:[0m kobj : 1.0
[1;38;5;39mCOMET INFO:[0m line_width : None
[1;38;5;39mCOMET INFO:[0m lr0 : 0.001
[1;38;5;39mCOMET INFO:[0m lrf : 0.01

```

📄[1;38;5;39mCOMET INFO:📄[0m	mask_ratio	: 4
📄[1;38;5;39mCOMET INFO:📄[0m	max_det	: 300
📄[1;38;5;39mCOMET INFO:📄[0m	mixup	: 0.0
📄[1;38;5;39mCOMET INFO:📄[0m	mode	: train
📄[1;38;5;39mCOMET INFO:📄[0m	model	: yolov8s.pt
📄[1;38;5;39mCOMET INFO:📄[0m	momentum	: 0.937
📄[1;38;5;39mCOMET INFO:📄[0m	mosaic	: 1.0
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📄[1;38;5;39mCOMET INFO:📄[0m	name	: yolov8s-2
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📄[1;38;5;39mCOMET INFO:📄[0m	optimizer	: AdamW
📄[1;38;5;39mCOMET INFO:📄[0m	overlap_mask	: True
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📄[1;38;5;39mCOMET INFO:📄[0m	perspective	: 0.0
📄[1;38;5;39mCOMET INFO:📄[0m	plots	: True

📄[1;38;5;39mCOMET INFO:📄[0m	pose	: 12.0
📄[1;38;5;39mCOMET INFO:📄[0m	pretrained	: True
📄[1;38;5;39mCOMET INFO:📄[0m	profile	: False
📄[1;38;5;39mCOMET INFO:📄[0m	project	: vehicle_runs
📄[1;38;5;39mCOMET INFO:📄[0m	quantize	: None
📄[1;38;5;39mCOMET INFO:📄[0m	rect	: False
📄[1;38;5;39mCOMET INFO:📄[0m	resume	: False
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📄[1;38;5;39mCOMET INFO:📄[0m	rle	: 1.0
📄[1;38;5;39mCOMET INFO:📄[0m	save	: True
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📄[1;38;5;39mCOMET INFO:📄[0m	save_crop	: False
📄[1;38;5;39mCOMET INFO:📄[0m	save_dir	: /content/runs/detect/vehicle_runs/yolov8s-2
📄[1;38;5;39mCOMET INFO:📄[0m	save_frames	: False
📄[1;38;5;39mCOMET INFO:📄[0m	save_json	: False
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📄[1;38;5;39mCOMET INFO:📄[0m	seed	: 0
📄[1;38;5;39mCOMET INFO:📄[0m	shear	: 0.0
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📄[1;38;5;39mCOMET INFO:📄[0m	warmup_momentum	: 0.8
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📄[1;38;5;39mCOMET INFO:📄[0m	Uploads:	
📄[1;38;5;39mCOMET INFO:📄[0m	asset	: 2 (3.72 KB)
📄[1;38;5;39mCOMET INFO:📄[0m	confusion-matrix	: 1
📄[1;38;5;39mCOMET INFO:📄[0m	environment details	: 1
📄[1;38;5;39mCOMET INFO:📄[0m	filename	: 1
📄[1;38;5;39mCOMET INFO:📄[0m	images	: 12
📄[1;38;5;39mCOMET INFO:📄[0m	installed packages	: 1
📄[1;38;5;39mCOMET INFO:📄[0m	model graph	: 1
📄[1;38;5;39mCOMET INFO:📄[0m	model-element	: 1 (21.46 MB)
📄[1;38;5;39mCOMET INFO:📄[0m	notebook	: 2
📄[1;38;5;39mCOMET INFO:📄[0m	os packages	: 1

[illegible]

```
[1;38;5;214mCOMET WARNING:[0m As you are running in a Jupyter environment, you will need to call 'experiment.end()' when finished to ensure all metrics and
[1;38;5;39mCOMET INFO:[0m Couldn't find a Git repository in '/content' nor in any parent directory. Set 'COMET_GIT_DIRECTORY' if your Git Repository is else
[1;38;5;39mCOMET INFO:[0m Experiment is live on comet.com https://www.comet.com/sarah-mohammadi/vehicle-yolov8/c96b57479c74ee58e4202064aad8879
```

Overriding model.yaml nc=80 with nc=5

	from	n	params	module	arguments
0	-1	1	1392	ultralitics.nn.modules.conv.Conv	[3, 48, 3, 2]
1	-1	1	41664	ultralitics.nn.modules.conv.Conv	[48, 96, 3, 2]
2	-1	2	111360	ultralitics.nn.modules.block.C2f	[96, 96, 2, True]
3	-1	1	166272	ultralitics.nn.modules.conv.Conv	[96, 192, 3, 2]
4	-1	4	813312	ultralitics.nn.modules.block.C2f	[192, 192, 4, True]
5	-1	1	664320	ultralitics.nn.modules.conv.Conv	[192, 384, 3, 2]
6	-1	4	3248640	ultralitics.nn.modules.block.C2f	[384, 384, 4, True]
7	-1	1	1991808	ultralitics.nn.modules.conv.Conv	[384, 576, 3, 2]
8	-1	2	3985920	ultralitics.nn.modules.block.C2f	[576, 576, 2, True]
9	-1	1	831168	ultralitics.nn.modules.block.SPPF	[576, 576, 5]

10	-1 1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
11	[-1, 6] 1	0	ultralalytics.nn.modules.conv.Concat	[1]
12	-1 2	1993728	ultralalytics.nn.modules.block.C2f	[960, 384, 2]
13	-1 1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
14	[-1, 4] 1	0	ultralalytics.nn.modules.conv.Concat	[1]
15	-1 2	517632	ultralalytics.nn.modules.block.C2f	[576, 192, 2]
16	-1 1	332160	ultralalytics.nn.modules.conv.Conv	[192, 192, 3, 2]
17	[-1, 12] 1	0	ultralalytics.nn.modules.conv.Concat	[1]
18	-1 2	1846272	ultralalytics.nn.modules.block.C2f	[576, 384, 2]
19	-1 1	1327872	ultralalytics.nn.modules.conv.Conv	[384, 384, 3, 2]
20	[-1, 9] 1	0	ultralalytics.nn.modules.conv.Concat	[1]
21	-1 2	4207104	ultralalytics.nn.modules.block.C2f	[960, 576, 2]
22	[15, 18, 21] 1	3778591	ultralalytics.nn.modules.head.Detect	[5, 16, None, [192, 384, 576]]
Model summary: 170 layers, 25,859,215 parameters, 25,859,199 gradients, 79.1 GFLOPs				
Remapped 5/5 cls head rows from pretrained weights by class name				
Transferred 475/475 items from pretrained weights				
Freezing layer 'model.22.dfl.conv.weight'				

[illegible][illegible]

		all	15	95	0.00136	0.05	0.000341	8.78e-05	
	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size		
图[K	13/60	3.466	0.7296	0.9892	1.079	63	640: 100% 0.000341 8.78e-05	9/9	3.6it/s 2.5s
图[K		Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 0.000341 8.78e-05	1/1	2.8it/s 0.4s
		all	15	95	0.508	0.194	0.00356 0.00246		
	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size		
图[K	14/60	3.466	0.7206	0.8848	1.027	79	640: 100% 0.000341 8.78e-05	9/9	3.7it/s 2.4s
图[K		Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 0.000341 8.78e-05	1/1	3.1it/s 0.3s
		all	15	95	0.778	0.103	0.0248 0.0189		
	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size		
图[K	15/60	3.466	0.7494	0.9252	1.046	62	640: 100% 0.000341 8.78e-05	9/9	3.4it/s 2.7s
图[K		Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 0.000341 8.78e-05	1/1	2.4it/s 0.4s
		all	15	95	0.783	0.111	0.034 0.0257		
	Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size		

```
Validating /content/runs/detect/vehicle_runs/yolo8m/weights/best.pt...  
Ultralytics 8.4.120 ðŸ‰ Python-3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)  
Model summary (fused): 93 layers, 25,842,655 parameters, 0 gradients, 78.7 GFLOPs  


|  | K   | Class | Images | Instances | Box(P) | R     | mAP50 | mAP50-95: | 100% | a" | b" | c" | d" | e" | f" | g" | h" | i" | j" | k" | l" | m" | n" | o" | p" | q" | r" | s" | t" | u" | v" | w" | x" | y"               | z" | Total |
|--|-----|-------|--------|-----------|--------|-------|-------|-----------|------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------------------|----|-------|
|  | all |       | 15     | 95        | 0.754  | 0.555 | 0.818 | 0.495     |      |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 1/1 2.7it/s 0.4s |    |       |

  
Speed: 0.1ms preprocess, 9.3ms inference, 0.0ms loss, 1.0ms postprocess per image  
Results saved to /content/runs/detect/vehicle_runs/yolo8mðŸŽ©
```

```

[1;38;5;39mCOMET INFO:[0m -----
[1;38;5;39mCOMET INFO:[0m Comet.ml Experiment Summary
[1;38;5;39mCOMET INFO:[0m -----
[1;38;5;39mCOMET INFO:[0m Data:
[1;38;5;39mCOMET INFO:[0m     display_summary_level : 1
[1;38;5;39mCOMET INFO:[0m     name                   : strange_dowel_6589
[1;38;5;39mCOMET INFO:[0m     url                    : https://www.comet.com/sarah-mohammadi/vehicle-yolov8/c96eb57479c74ee58e4202064aad8879
[1;38;5;39mCOMET INFO:[0m Metrics [count] (min, max):
[1;38;5;39mCOMET INFO:[0m     loss [13]              : (21025.5078125, 1521046.0)
[1;38;5;39mCOMET INFO:[0m     lr/pg0 [17]            : (0.0002962962962962963, 0.0009500000000000001)
[1;38;5;39mCOMET INFO:[0m     lr/pg1 [17]            : (0.0002962962962962963, 0.0009500000000000001)
[1;38;5;39mCOMET INFO:[0m     lr/pg2 [17]            : (0.0007524999999999999, 0.07066666666666667)
[1;38;5;39mCOMET INFO:[0m     metrics/mAP50(B) [17]  : (2e-05, 0.81849)
[1;38;5;39mCOMET INFO:[0m     metrics/mAP50-95(B) [17] : (1e-05, 0.52059)
[1;38;5;39mCOMET INFO:[0m     metrics/precision(B) [17] : (0.00034, 0.8199)
[1;38;5;39mCOMET INFO:[0m     metrics/recall(B) [17]  : (0.01111, 0.55556)
[1;38;5;39mCOMET INFO:[0m     model/gFLOPs           : 79.072
[1;38;5;39mCOMET INFO:[0m     model/parameters       : 25859215

```

```
[1;38;5;39mCOMET INFO:[0m model/speed_PyTorch(ms) : 8.476
[1;38;5;39mCOMET INFO:[0m train/box_loss [16] : (0.56986, 1.45651)
[1;38;5;39mCOMET INFO:[0m train/cls_loss [16] : (0.79467, 3.84716)
[1;38;5;39mCOMET INFO:[0m train/df1_loss [16] : (0.95743, 1.45358)
[1;38;5;39mCOMET INFO:[0m val/box_loss [16] : (0.80955, 3.15061)
[1;38;5;39mCOMET INFO:[0m val/cls_loss [16] : (2.12017, 6610.88184)
[1;38;5;39mCOMET INFO:[0m val/df1_loss [16] : (1.05807, 10.51215)
Others:
[1;38;5;39mCOMET INFO:[0m eval_batch_logging_interval : 1
[1;38;5;39mCOMET INFO:[0m log_confusion_matrix_on_eval : False
[1;38;5;39mCOMET INFO:[0m log_image_predictions : True
[1;38;5;39mCOMET INFO:[0m max_image_predictions : 100
[1;38;5;39mCOMET INFO:[0m notebook_url : https://colab.research.google.com/notebook#fileId=1LLavEZtYhkYpjPRrvtMnhSrtv2fne9v
Parameters:
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[1;38;5;39mCOMET INFO:[0m amp : True
[1;38;5;39mCOMET INFO:[0m angle : 1.0
[1;38;5;39mCOMET INFO:[0m augment : False
```

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0[1,38;5,39mCOMET INFO:0[0m box : 7.5
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0[1,38;5,39mCOMET INFO:0[0m cfg : None
0[1,38;5,39mCOMET INFO:0[0m channels_last : False
0[1,38;5,39mCOMET INFO:0[0m classes : None
0[1,38;5,39mCOMET INFO:0[0m close_mosaic : 10
0[1,38;5,39mCOMET INFO:0[0m cls : 0.5
0[1,38;5,39mCOMET INFO:0[0m cls_pw : 0.0
0[1,38;5,39mCOMET INFO:0[0m cls_remap : True
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0[1,38;5,39mCOMET INFO:0[0m copy_paste_mode : flip
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0[1,38;5,39mCOMET INFO:0[0m cutmix :
```

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1;38;5;39mCOMET INFO:0m	dfl	: 1.5
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1;38;5;39mCOMET INFO:0m	hsv_s	: 0.7
1;38;5;39mCOMET INFO:0m	hsv_v	: 0.4
1;38;5;39mCOMET INFO:0m	ingsz	: 640
1;38;5;39mCOMET INFO:0m	iou	: 0.7
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1;38;5;39mCOMET INFO:0m	kobj	: 1.0
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1;38;5;39mCOMET INFO:0m	lrf	: 0.01
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1;38;5;39mCOMET INFO:0m	max_det	: 300
1;38;5;39mCOMET INFO:0m	mixup	: 0.0

1;38;5;39mCOMET INFO:0m	mode	: train
1;38;5;39mCOMET INFO:0m	model	: yolov8m.pt
1;38;5;39mCOMET INFO:0m	momentum	: 0.937
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1;38;5;39mCOMET INFO:0m	optimizer	: AdamW
1;38;5;39mCOMET INFO:0m	overlap_mask	: True
1;38;5;39mCOMET INFO:0m	patience	: 15
1;38;5;39mCOMET INFO:0m	perspective	: 0.0
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1;38;5;39mCOMET INFO:0m	resume	: False
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1;38;5;39mCOMET INFO:0m	scale	: 0.5
1;38;5;39mCOMET INFO:0m	seed	: 0
1;38;5;39mCOMET INFO:0m	shear	: 0.0
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```

[1,38;5;39mCOMET INFO:[0m show_boxes : True
[1,38;5;39mCOMET INFO:[0m show_conf : True
[1,38;5;39mCOMET INFO:[0m show_labels : True
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[1,38;5;39mCOMET INFO:[0m single_cls : False
[1,38;5;39mCOMET INFO:[0m source : None
[1,38;5;39mCOMET INFO:[0m split : val
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[1,38;5;39mCOMET INFO:[0m task : detect
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[1,38;5;39mCOMET INFO:[0m tracker : tracktrack.yaml
[1,38;5;39mCOMET INFO:[0m translate : 0.1
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[1,38;5;39mCOMET INFO:[0m warmup_bias_lr : 0.1
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```

```
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[1;38;5;39mCOMET INFO:[0m      workspace : None
Uploads:
[1;38;5;39mCOMET INFO:[0m      asset : 2 (3.71 KB)
[1;38;5;39mCOMET INFO:[0m      confusion-matrix : 1
[1;38;5;39mCOMET INFO:[0m      environment details : 1
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[1;38;5;39mCOMET INFO:[0m      images : 12
[1;38;5;39mCOMET INFO:[0m      installed packages : 1
[1;38;5;39mCOMET INFO:[0m      model graph : 1
[1;38;5;39mCOMET INFO:[0m      model-element : 1 (49.60 MB)
[1;38;5;39mCOMET INFO:[0m      notebook : 2
[1;38;5;39mCOMET INFO:[0m      os packages : 1
[1;38;5;39mCOMET INFO:[0m      source_code : 1
[1;38;5;39mCOMET INFO:[0m
[1;38;5;39mCOMET INFO:[0m Please wait for assets to finish uploading (timeout is 10800 seconds)
```

[illegible]

```

[1]38;5;39mCOMET INFO: Couldnt find a Git repository in '/content' nor in any parent directory. Set `COMET_GIT_DIRECTORY` if your Git Repository is els
Overriding model.yaml nc=80 with nc=5

```

	from	params	module	arguments
0	-1 1	1856	ultralytics.nn.modules.conv.Conv	[3, 64, 3, 2]
1	-1 1	73984	ultralytics.nn.modules.conv.Conv	[64, 128, 3, 2]
2	-1 3	279808	ultralytics.nn.modules.block.C2f	[128, 128, 3, True]
3	-1 1	295424	ultralytics.nn.modules.conv.Conv	[128, 256, 3, 2]
4	-1 6	2101248	ultralytics.nn.modules.block.C2f	[256, 256, 6, True]
5	-1 1	1180672	ultralytics.nn.modules.conv.Conv	[256, 512, 3, 2]
6	-1 6	8396800	ultralytics.nn.modules.block.C2f	[512, 512, 6, True]
7	-1 1	2360320	ultralytics.nn.modules.conv.Conv	[512, 512, 3, 2]
8	-1 3	4461568	ultralytics.nn.modules.block.C2f	[512, 512, 3, True]
9	-1 1	656896	ultralytics.nn.modules.block.SPPF	[512, 512, 5]
10	-1 1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
11	[-1, 6] 1	0	ultralytics.nn.modules.conv.Concat	[1]

```
Remapped 5/5 cls head rows from pretrained weights by class name
Transferred 595/595 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
[34m[1mAMP: @0mrunning Automatic Mixed Precision (AMP) checks...
[34m[1mAMP: @0mchecks passed â€¦
```

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
0[K	1/60	5.07G	0.5428	1.34	0.9626	96
0[K		Class	Images	Instances	Box(P	R
		all	15	95	0.585	0.464
						0.407
						0.247
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size

[illegible]

[illegible]

```

all      15      95      0.551      0.0886      0.0316      0.0169

Epoch GPU_mem box_loss cls_loss dfl_loss Instances Size
[0]K    17/60   5.51G   1.106   1.371   1.355   80   640: 100% 1/1 2.3it/s 3.8s
[0]K          Class Images Instances Box(P R mAP50 mAP50-95): 100% 1/1 2.1it/s 0.5s
all      15      95      0.53      0.0917      0.0155      0.0076

[0]34mEarlyStopping: Training stopped early as no improvement observed in last 15 epochs. Best results observed at epoch 2, best model saved as best.pt
To update EarlyStopping(patience=15) pass a new patience value, i.e. `patience=300` or use `patience=0` to disable EarlyStopping.

17 epochs completed in 0.033 hours.
Optimizer stripped from /content/runs/detect/vehicle_runs/yolov8l/weights/last.pt, 87.6MB
Optimizer stripped from /content/runs/detect/vehicle_runs/yolov8l/weights/best.pt, 87.6MB

Validating /content/runs/detect/vehicle_runs/yolov8l/weights/best.pt...
Ultralytics 8.4.120 Python 3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
Model summary (fused): 113 layers, 43,610,463 parameters, 0 gradients, 164.8 GFLOPs
[0]K          Class Images Instances Box(P R mAP50 mAP50-95): 100% 1/1 2.3it/s 0.4s
all      15      95      0.856      0.464      0.434      0.268
```

```
Speed: 0.1ms preprocess, 14.0ms inference, 0.0ms loss, 1.0ms postprocess per image
Results saved to [1m/content/runs/detect/vehicle_runs/yolov8l[0m

[1;38;5;39mCOMET INFO:[0m -----
[1;38;5;39mCOMET INFO:[0m Comet.ml Experiment Summary
[1;38;5;39mCOMET INFO:[0m -----
[1;38;5;39mCOMET INFO:[0m Data:
[1;38;5;39mCOMET INFO:[0m     display_summary_level : 1
[1;38;5;39mCOMET INFO:[0m     name                  : gentle_supernova_7283
[1;38;5;39mCOMET INFO:[0m     url                   : https://www.comet.com/sarah-mohammadi/vehicle-yolov8/bac2458f42cc4316a016027362f34c68
[1;38;5;39mCOMET INFO:[0m Metrics [count] (min, max):
[1;38;5;39mCOMET INFO:[0m     loss [14]              : (7690.533203125, 1320648.0)
[1;38;5;39mCOMET INFO:[0m     lr/pg0 [18]            : (0.0002962962962962963, 0.0009505000000000001)
[1;38;5;39mCOMET INFO:[0m     lr/pg1 [18]            : (0.0002962962962962963, 0.0009505000000000001)
[1;38;5;39mCOMET INFO:[0m     lr/pg2 [18]            : (0.0007360000000000001, 0.07066666666666667)
[1;38;5;39mCOMET INFO:[0m     metrics/mAP50(B) [18]  : (0.0, 0.43609)
[1;38;5;39mCOMET INFO:[0m     metrics/mAP50-95(B) [18] : (0.0, 0.26973)
[1;38;5;39mCOMET INFO:[0m     metrics/precision(B) [18] : (0.0, 0.88043)
```

```

[1;38;5;39mCOMET INFO:[0m metrics/recall(B) [18] : (0.0, 0.46389)
[1;38;5;39mCOMET INFO:[0m model/GFLOPs : 165.412
[1;38;5;39mCOMET INFO:[0m model/parameters : 43633695
[1;38;5;39mCOMET INFO:[0m model/speed_PyTorch(ms) : 12.601
[1;38;5;39mCOMET INFO:[0m train/box_loss [17] : (0.54279, 2.77374)
[1;38;5;39mCOMET INFO:[0m train/cls_loss [17] : (1.33976, 5.99485)
[1;38;5;39mCOMET INFO:[0m train/df1_loss [17] : (0.96262, 2.41661)
[1;38;5;39mCOMET INFO:[0m val/box_loss [17] : (1.23076, 3.15634)
[1;38;5;39mCOMET INFO:[0m val/cls_loss [17] : (3.98154, 826.3811)
[1;38;5;39mCOMET INFO:[0m val/df1_loss [17] : (1.34141, 3.37671)
[1;38;5;39mCOMET INFO:[0m Others:
[1;38;5;39mCOMET INFO:[0m eval_batch_logging_interval : 1
[1;38;5;39mCOMET INFO:[0m log_confusion_matrix_on_eval : False
[1;38;5;39mCOMET INFO:[0m log_image_predictions : True
[1;38;5;39mCOMET INFO:[0m max_image_predictions : 100
[1;38;5;39mCOMET INFO:[0m notebook_url : https://colab.research.google.com/notebook#fileId=1ILLavEZtYHkYpJPRrvtMnhSrtv2fne9v
[1;38;5;39mCOMET INFO:[0m Parameters:
[1;38;5;39mCOMET INFO:[0m agnostic_nms : False

```

```

[1;38;5;39mCOMET INFO:[0m amp : True
[1;38;5;39mCOMET INFO:[0m angle : 1.0
[1;38;5;39mCOMET INFO:[0m augment : False
[1;38;5;39mCOMET INFO:[0m auto_augment : randaugment
[1;38;5;39mCOMET INFO:[0m batch : 8
[1;38;5;39mCOMET INFO:[0m bgr : 0.0
[1;38;5;39mCOMET INFO:[0m box : 7.5
[1;38;5;39mCOMET INFO:[0m cache : False
[1;38;5;39mCOMET INFO:[0m cfg : None
[1;38;5;39mCOMET INFO:[0m channels_last : False
[1;38;5;39mCOMET INFO:[0m classes : None
[1;38;5;39mCOMET INFO:[0m close_mosaic : 10
[1;38;5;39mCOMET INFO:[0m cls : 0.5
[1;38;5;39mCOMET INFO:[0m cls_pw : 0.0
[1;38;5;39mCOMET INFO:[0m cls_remap : True
[1;38;5;39mCOMET INFO:[0m compile : False
[1;38;5;39mCOMET INFO:[0m conf : None
[1;38;5;39mCOMET INFO:[0m copy_paste : 0.0

```

```

[1;38;5;39mCOMET INFO:[0m copy_paste_mode : flip
[1;38;5;39mCOMET INFO:[0m cos_lr : False
[1;38;5;39mCOMET INFO:[0m cutmix : 0.0
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[1;38;5;39mCOMET INFO:[0m degrees : 0.0
[1;38;5;39mCOMET INFO:[0m deterministic : True
[1;38;5;39mCOMET INFO:[0m device : 0
[1;38;5;39mCOMET INFO:[0m df1 : 1.5
[1;38;5;39mCOMET INFO:[0m dgrad : 0.5
[1;38;5;39mCOMET INFO:[0m dis : 6.0
[1;38;5;39mCOMET INFO:[0m distill_model : None
[1;38;5;39mCOMET INFO:[0m dlam : 1.0
[1;38;5;39mCOMET INFO:[0m dlog : 1.0
[1;38;5;39mCOMET INFO:[0m dnn : False
[1;38;5;39mCOMET INFO:[0m dropout : 0.0
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[1;38;5;39mCOMET INFO:[0m embed : None
[1;38;5;39mCOMET INFO:[0m end2end : None

```

```

[1;38;5;39mCOMET INFO:[0m epochs : 60
[1;38;5;39mCOMET INFO:[0m erasing : 0.4
[1;38;5;39mCOMET INFO:[0m exist_ok : False
[1;38;5;39mCOMET INFO:[0m fliplr : 0.5
[1;38;5;39mCOMET INFO:[0m flipud : 0.0
[1;38;5;39mCOMET INFO:[0m format : torchscript
[1;38;5;39mCOMET INFO:[0m fraction : 1.0
[1;38;5;39mCOMET INFO:[0m freeze : None
[1;38;5;39mCOMET INFO:[0m hsv_h : 0.015
[1;38;5;39mCOMET INFO:[0m hsv_s : 0.7
[1;38;5;39mCOMET INFO:[0m hsv_v : 0.4
[1;38;5;39mCOMET INFO:[0m imsz : 640
[1;38;5;39mCOMET INFO:[0m iou : 0.7
[1;38;5;39mCOMET INFO:[0m keras : False
[1;38;5;39mCOMET INFO:[0m kobj : 1.0
[1;38;5;39mCOMET INFO:[0m line_width : None
[1;38;5;39mCOMET INFO:[0m lr0 : 0.001
[1;38;5;39mCOMET INFO:[0m lrf : 0.01

```



```

[1;38;5;39mCOMET INFO:[0m      mask_ratio      : 4
[1;38;5;39mCOMET INFO:[0m      max_det       : 300
[1;38;5;39mCOMET INFO:[0m      mixup         : 0.0
[1;38;5;39mCOMET INFO:[0m      mode          : train
[1;38;5;39mCOMET INFO:[0m      model         : yolov8l.pt
[1;38;5;39mCOMET INFO:[0m      momentum     : 0.937
[1;38;5;39mCOMET INFO:[0m      mosaic        : 1.0
[1;38;5;39mCOMET INFO:[0m      multi_scale   : 0.0
[1;38;5;39mCOMET INFO:[0m      name          : yolov8l
[1;38;5;39mCOMET INFO:[0m      nbs           : 64
[1;38;5;39mCOMET INFO:[0m      nms           : False
[1;38;5;39mCOMET INFO:[0m      opset         : None
[1;38;5;39mCOMET INFO:[0m      optimize      : False
[1;38;5;39mCOMET INFO:[0m      optimizer     : AdamW
[1;38;5;39mCOMET INFO:[0m      overlap_mask  : True
[1;38;5;39mCOMET INFO:[0m      patience      : 15
[1;38;5;39mCOMET INFO:[0m      perspective    : 0.0
[1;38;5;39mCOMET INFO:[0m      plots         : True

```

```

[1;38;5;39mCOMET INFO:[0m      pose          : 12.0
[1;38;5;39mCOMET INFO:[0m      pretrained    : True
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[1;38;5;39mCOMET INFO:[0m      project       : vehicle_runs
[1;38;5;39mCOMET INFO:[0m      quantize      : None
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[1;38;5;39mCOMET INFO:[0m      resume        : False
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[1;38;5;39mCOMET INFO:[0m      rle           : 1.0
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[1;38;5;39mCOMET INFO:[0m      save_conf     : False
[1;38;5;39mCOMET INFO:[0m      save_crop     : False
[1;38;5;39mCOMET INFO:[0m      save_dir      : /content/runs/detect/vehicle_runs/yolov8l
[1;38;5;39mCOMET INFO:[0m      save_frames   : False
[1;38;5;39mCOMET INFO:[0m      save_json     : False
[1;38;5;39mCOMET INFO:[0m      save_period   : -1
[1;38;5;39mCOMET INFO:[0m      save_txt      : False
[1;38;5;39mCOMET INFO:[0m      scale         : 0.5

```

```

[1;38;5;39mCOMET INFO:[0m      seed          : 0
[1;38;5;39mCOMET INFO:[0m      shear         : 0.0
[1;38;5;39mCOMET INFO:[0m      show          : False
[1;38;5;39mCOMET INFO:[0m      show_boxes    : True
[1;38;5;39mCOMET INFO:[0m      show_conf     : True
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[1;38;5;39mCOMET INFO:[0m      simplify      : True
[1;38;5;39mCOMET INFO:[0m      single_cls    : False
[1;38;5;39mCOMET INFO:[0m      source        : None
[1;38;5;39mCOMET INFO:[0m      split         : val
[1;38;5;39mCOMET INFO:[0m      stream_buffer  : False
[1;38;5;39mCOMET INFO:[0m      task          : detect
[1;38;5;39mCOMET INFO:[0m      time          : None
[1;38;5;39mCOMET INFO:[0m      tracker       : tracktrack.yaml
[1;38;5;39mCOMET INFO:[0m      translate     : 0.1
[1;38;5;39mCOMET INFO:[0m      val           : True
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[1;38;5;39mCOMET INFO:[0m      vid_stride    : 1

```

```

[1;38;5;39mCOMET INFO:[0m      visualize     : False
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[1;38;5;39mCOMET INFO:[0m      warmup_epochs  : 3.0
[1;38;5;39mCOMET INFO:[0m      warmup_momentum : 0.8
[1;38;5;39mCOMET INFO:[0m      weight_decay   : 0.0005
[1;38;5;39mCOMET INFO:[0m      workers        : 8
[1;38;5;39mCOMET INFO:[0m      workspace      : None
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[1;38;5;39mCOMET INFO:[0m      asset          : 2 (3.50 KB)
[1;38;5;39mCOMET INFO:[0m      confusion-matrix : 1
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[1;38;5;39mCOMET INFO:[0m      installed packages : 1
[1;38;5;39mCOMET INFO:[0m      model graph     : 1
[1;38;5;39mCOMET INFO:[0m      model-element   : 1 (83.57 MB)
[1;38;5;39mCOMET INFO:[0m      notebook       : 2
[1;38;5;39mCOMET INFO:[0m      os packages     : 1

```


[illegible]

سه مدل کوچک، متوسط و بزرگ طی ۶۰ دوره آموزش می‌بینند. خروجی کامل آموزش و ارزیابی هر مدل، مسیر بهترین وزن‌ها و معیارهای Precision، Recall، mAP50 و mAP50-95 درج شده است.

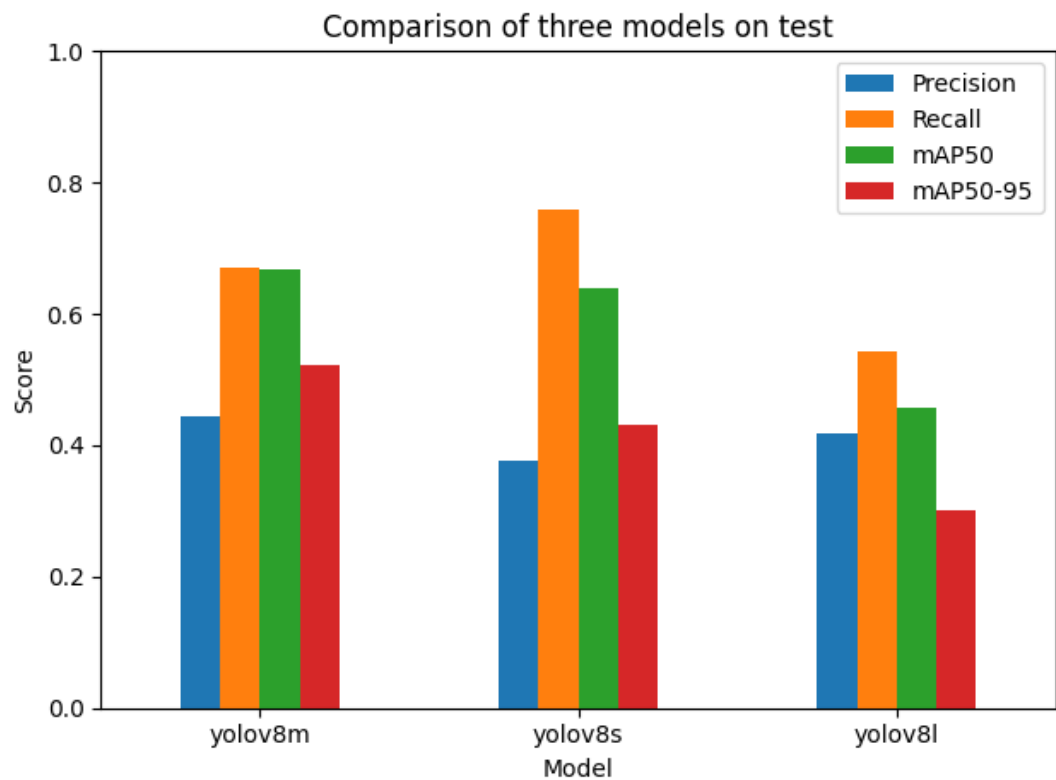
۱۰. جدول و نمودار مقایسه مدل‌ها

```
comparison_df = pd.DataFrame(results_rows).sort_values('mAP50-95', ascending=False)
display(comparison_df)

comparison_df.set_index('Model')[['Precision', 'Recall', 'mAP50', 'mAP50-95']].plot(kind='bar', ylim=(0, 1), rot=0)
plt.ylabel('Score')
plt.title('Comparison of three models on test')
plt.tight_layout()
plt.savefig('model_comparison.png', dpi=150)
plt.show()
```

Model	Precision	Recall	mAP50	mAP50-95	Inference_ms \
1 yolov8m	0.444172	0.671264	0.668539	0.521615	17.596550
0 yolov8s	0.377198	0.759416	0.638866	0.432415	7.968208
2 yolov8l	0.417190	0.542529	0.456945	0.302063	27.455777

	Train_minutes
1	1.432281
0	1.135375
2	2.493145



جدول مقایسه بر اساس mAP50-95 مرتب شده و نمودار میله‌ای چهار معیار اصلی را برای سه مدل نمایش می‌دهد.

۱۱. نمایش پیش‌بینی بهترین مدل

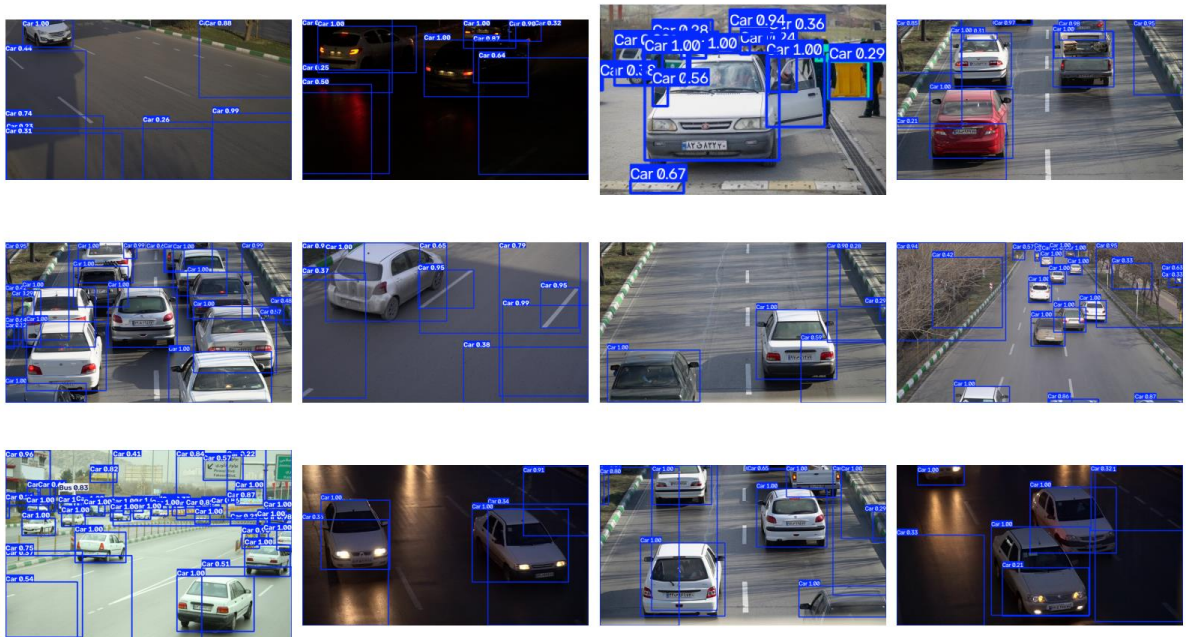
```
best_name = comparison_df.iloc[0]['Model']
best_model = YOLO(model_paths[best_name]['best'])

predictions = best_model.predict(
    source=str(DATASET_DIR / 'images' / 'test'), conf=0.2, iou=0.5, imgsiz=IMG_SIZE,
    device=DEVICE, save=True, verbose=False
)

fig, axes = plt.subplots(3, 4, figsize=(16, 10))
for ax, result in zip(axes.ravel(), predictions[:12]):
    ax.imshow(cv2.cvtColor(result.plot(), cv2.COLOR_BGR2RGB))
    ax.axis('off')
plt.suptitle(f'Model prediction{best_name} on test')
plt.tight_layout()
plt.savefig('best_model_predictions.jpg', dpi=150)
plt.show()
```

Results saved to @1m/content/runs/detect/predict@0m

Model predictionyolov8m on test

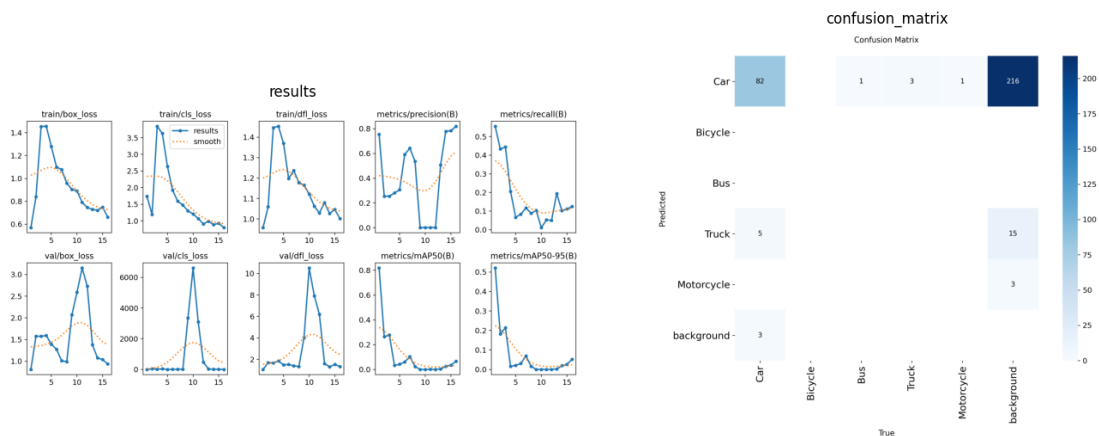


مدل دارای بالاترین mAP50-95 انتخاب شده و پیش‌بینی آن روی دوازده تصویر آزمون نمایش داده می‌شود.

۱۲. نمودارهای تشخیصی آموزش

```
train_dir = Path(model_paths[best_name]['train_dir'])
diagnostic_files = [train_dir / f for f in ['results.png', 'confusion_matrix.png', 'PR_curve.png'] if (train_dir / f).exists()]

fig, axes = plt.subplots(1, len(diagnostic_files), figsize=(7 * len(diagnostic_files), 6))
axes = np.atleast_1d(axes)
for ax, path in zip(axes, diagnostic_files):
    ax.imshow(PILImage.open(path))
    ax.set_title(path.stem)
    ax.axis('off')
plt.tight_layout()
plt.show()
```



نمودار نتایج آموزش، ماتریس درهم‌ریختگی و منحنی Precision-Recall بهترین مدل برای تحلیل نهایی کنار هم قرار گرفته‌اند.

جمع‌بندی نتایج

فرایند کامل از دریافت تصاویر تا برچسب‌گذاری خودکار، تقسیم داده، آموزش، مقایسه و نمایش پیش‌بینی‌ها اجرا شده است. استفاده از YOLOv5 برای تولید برچسب‌های اولیه و پردازش جریانی، مشکل کمبود حافظه مشاهده‌شده در اجرای قبلی را کاهش می‌دهد. نتایج ذخیره‌شده نشان می‌دهند افزایش اندازه مدل لزوماً بدون هزینه نیست و انتخاب نهایی باید بر اساس توازن mAP50-95، زمان آموزش و منابع GPU انجام شود.